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Electricity demand forecasting at distribution and household levels using explainable causal graph neural network

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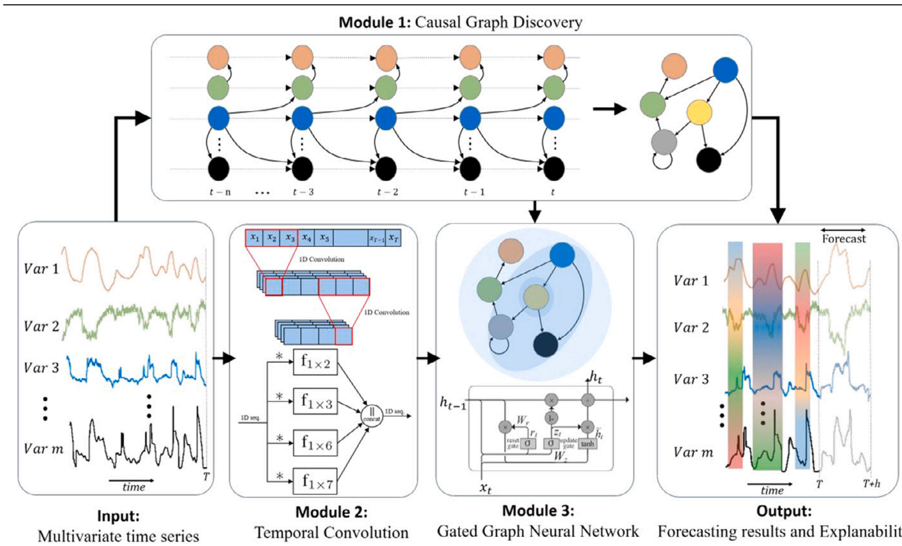
HIGHLIGHTS

- **Innovative Forecasting Approach:** Introduces eXplainable Causal Graph Neural Network (X-CGNN) for electricity demand forecasting, addressing challenges posed by black-box models prevalent in deep neural networks.
- **Multivariate Forecasting Capability:** Overcomes limitations of existing methods by capturing inter-dependencies among variables, especially crucial in the context of multivariate time series forecasting.
- **Dual Explanatory Framework:** Provides intrinsic and global explanations based on causal inferences, offering insights into the relationships between variables. Additionally, incorporates local explanations through post-hoc analyses for a comprehensive understanding.
- **Performance Advancements:** Demonstrates state-of-the-art performance through extensive validation on real-world electricity demand datasets at both household and distribution levels. The proposed framework excels in both short-term and long-term demand forecasting.
- **Enhanced Graph Neural Network Performance:** Establishes meaningful neighborhoods based on causality, enhancing the performance of Graph Neural Network structures. The framework achieves improved predictive accuracy by incorporating temporal features and causal interdependencies in electricity demand prediction.

GRAPHICAL ABSTRACT



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ABSTRACT

Forecasting electricity demand is an essential part of the smart grid to ensure a stable and reliable power grid. With the increasing integration of renewable energy resources into the grid, forecasting the demand for electricity is critical at all levels, from the distribution to the household. Most existing forecasting methods, however, can be considered black-box models as a result of deep digitalization enablers, such as deep neural networks, which remain difficult to interpret by humans. Moreover, capture of the inter-dependencies among variables presents a significant challenge for multivariate time series forecasting. In this paper we propose eXplainable Causal Graph Neural Network (X-CGNN) for multivariate electricity demand forecasting that overcomes these limitations. As part of this method, we have intrinsic and global explanations based on causal inferences as well as local explanations based on post-hoc analyses. We have performed extensive validation on two real-world electricity demand datasets from both the household and distribution levels to demonstrate that our proposed method achieves state-of-the-art performance.

1. Introduction

Electricity demand forecasting is a critical part of the planning process in the electricity industry [1] and a critical part of the operation of electric power systems [2]. Forecasting the electricity demand is closely related to the development of the economy, as well as to national security and societal functioning. In this regard, accurate electric demand forecasting is of great importance for planning energy generating capacity and managing power systems [3], since it results in substantial reductions in operating and maintenance costs as well as correct decisions for future development [4].

Electricity demand data are inherently time series. Methods for predicting time series data have been employed for predicting electricity demand, and they can generally be divided into two categories: statistical methods and machine learning methods. statistical methods such as AutoRegressive (AR) [5,6], ARIMA [7] and exponential smoothing [8] models are used to predict electricity demand. These statistical methods for electricity demand forecasting have limitations that affect their effectiveness. They struggle with capturing complex and nonlinear demand patterns, are sensitive to data quality issues, and face challenges when dealing with high-dimensional data.

Machine learning models have recently enhanced the predictive capabilities for electricity demand forecasting and solved these limitations [9–11]. Commonly used machine learning models for predicting electricity demand data in power system application include Support Vector Regression (SVR) [12], Random Forests (RF) [13], Recurrent Neural Networks (RNNs) [14] and Long Short-Term Memory (LSTM) [15]. Although machine learning methods have provided higher accuracy in prediction but a disadvantage of complex machine learning models is that they are impossible for even the most competent data scientists to understand, making it difficult for experts or end-user to trust them and apply them effectively. Additionally, the mentioned models may not fully capture the complex interdependencies and connections between variables.

In order to address these issues, we propose an innovative approach called **eXplainable Causal Graph Neural Network (X-CGNN)** for electricity demand forecasting. Inspired by recent works in causal inference [16,17], we build causal graphs from time series data to represent the relationships between variables. Also we use eXplainable Artificial Intelligence (XAI) [18] tools for local explanations of the proposed method. A brief overview of this study is shown in Fig. 1. The main contributions of our work are:

- We propose a framework for predicting electricity demand using temporal features and causal interdependencies. Our proposed framework offering two key advantages: (i) Enhanced GNN Performance: By establishing meaningful neighborhoods based on causality, we achieve improved performance for graph neural network structures. (ii) Global and Local Explanations: Our method provides both global and local explanations for the prediction process. Global explanations are derived from the causal graph, illustrating the causal relationships between variables. Local explanations, on the other hand, are generated using XAI tools, revealing the importance of each time lag in prediction and the impact of each variable on the prediction outcome.
- Extensive validation on two real-world electricity demand datasets from household level and also distribution level to demonstrate that our proposed method achieves state-of-the-art performance in both short-term and long-term electricity demand forecasting.

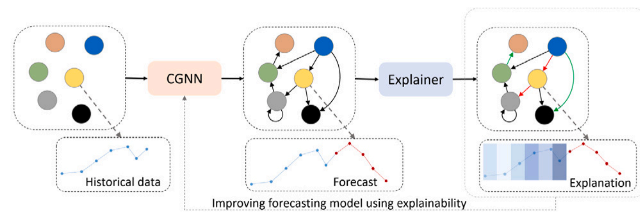


Fig. 1. Methodology overview of the proposed X-CGNN method.

2. Methodology

Fig. 2 provides a visual representation of the overall architecture of the proposed model, which is composed of three integral modules: the causal graph discovery (CGD), temporal convolution (TC), and graph neural network (GNN). The following sections provide details on these three major modules.

2.1. Module 1: Causal graph discovery

In order to comprehend large or complex datasets, graphs are a popular approach. Notably, as the demand for interpretable models and causal insights into underlying processes grows, directed acyclic graphs (DAGs) have emerged as promising tools in various applications. DAGs reveal the inter-relationships between variables in a system through their edges. In this study, we extract the causality graph from observed data.

Given a data matrix $\mathbf{X} \in \mathbb{R}^{T\times N}$ consisting of T observations of vector $X = \left(X_1, \dots, X_N\right)$ and a (discrete) space of DAGs denoted by $\mathbb{D}$, where $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a DAG with N nodes, our objective is to learn DAG $\mathcal{G} \in \mathbb{D}$ that models the joint distribution $\mathbb{P}(X)$ [19]. To accomplish this, we adopt a structural equation model (SEM) for X that is defined by a weighted adjacency matrix $W \in \mathbb{R}^{N\times N}$. Any $W \in \mathbb{R}^{N\times N}$ defines a graph on N nodes in the following way: Let $A(W) \in \{0,1\}^{N\times N}$ be the binary matrix such that $[A(W)]_{ij} = 1 \Leftrightarrow w_{ij} \neq 0$ and zero otherwise; then $A(W)$ defines the adjacency matrix of a directed graph $G(W)$. With considering the vectors of weighted adjacency matrix $W = \left[w_1 | \cdots | w_N\right]$ we can define SEM by $X_j = w_j^T X + z_j$ or in general form based on parent variable $\mathbb{E}\left(X_j \mid X_{pa}\left(X_j\right)\right) = f(w_j^T X)$, so we can model the

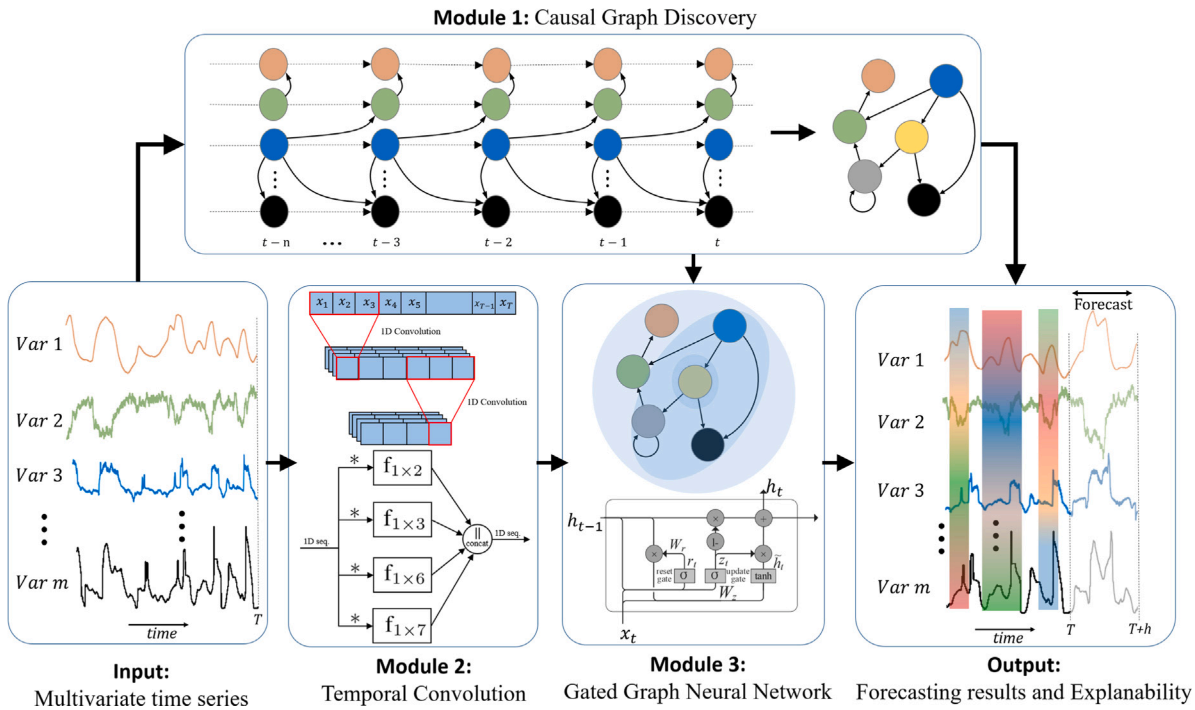


Fig. 2. Pipeline of the proposed approach. Module 1 constructs causality graphs that reveal hidden relationships between variables. In module 2, time features are extracted using dilated convolution on the time series. Based on the causality graph of the first module and the extracted temporal features of module 2, module 3 performs the prediction model by applying the GNN to the data and considering the causality graph.

conditional distribution of Xj given its parents. The general problem we are attempting to solve to determine the adjacency matrix is as follows:

$$\begin{aligned} &\min_{W\in\mathbb{R}^{N\times N}} F(W) \\ &\text{s.t} \quad G(W)\in\mathbb{D}. \end{aligned} \tag{1}$$

where $G(W)$ is the N-node graph induced by the weighted adjacency matrix W. We use least-squares (LS) loss $\ell(W;\mathbf{X})$ and ℓ_1-regularization $\|W\|_1 = \|\text{vec}(W)\|_1$ to define score function as:

$$F(W) = \ell(W;\mathbf{X}) + \lambda\,\|W\|_1 = \frac{1}{2N}\,\|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda\,\|W\|_1\,. \tag{2}$$

Our optimization problem (1) can be efficiently solved and we can identify a DAG that best represents inter-dependencies between variables using gradient-based numerical methods and continuous optimization techniques presented in [20]. In addition to serving as the underlying graph in the GNN module, this directed graph also provides a way to evaluate the interpretability of the mode.

2.2. Module 2: Temporal convolution

Temporal convolution involves sliding a fixed-size window (filter) over time series data to extract temporal patterns. Instead of conventional convolution, we use dilated convolution [21]. A dilated convolution is an efficient way of analyzing large-scale time data because it captures long-term dependencies in input sequences, which are essential for accurate prediction and analysis of time data.

For one-dimensional time series data of size T ($X \in \mathbb{R}^T$) and a filter of size K ($\mathrm{f}_{1\times K} \in \mathbb{R}^K$), dilated convolution define as follows:

$$X * \mathrm{f}_{1\times K}(t) = \sum_{s=0}^{K-1} \mathrm{f}_{1\times K}(s)X(t - d\times s), \tag{3}$$

where d is the dilation factor. In time series data, recurring patterns like daily, weekly, or hourly cycles are often present and cannot be accurately captured by a single filter. To address this, multiple filters with different time lengths are commonly used for temporal convolution to capture all relevant features [22]. In many cases, temporal signals exhibit multiple inherent periods. For our application, periods 7, 12, and 24 are naturally significant for energy demand. Using filter sizes of 1×2, 1×3, 1×6, and 1×7, we can accommodate all these periods through various combinations of these filter sizes. Thus in this approach, we heuristically selected filters with lengths of 2, 3, 6, and 7 time steps. The outputs of these filters are concatenated such that $\widetilde{X} = \text{concat}(X * \mathrm{f}_{1\times2}, X * \mathrm{f}_{1\times3}, X * \mathrm{f}_{1\times6}, X * \mathrm{f}_{1\times7})$, where X represents the one-dimensional time series input and $\widetilde{X}$ is the output of the convolution with four different-sized filters and concatenation. Fig. 2 (module 2) illustrates the concept of time convolution as discussed in this section.

2.3. Module 3: Graph neural network

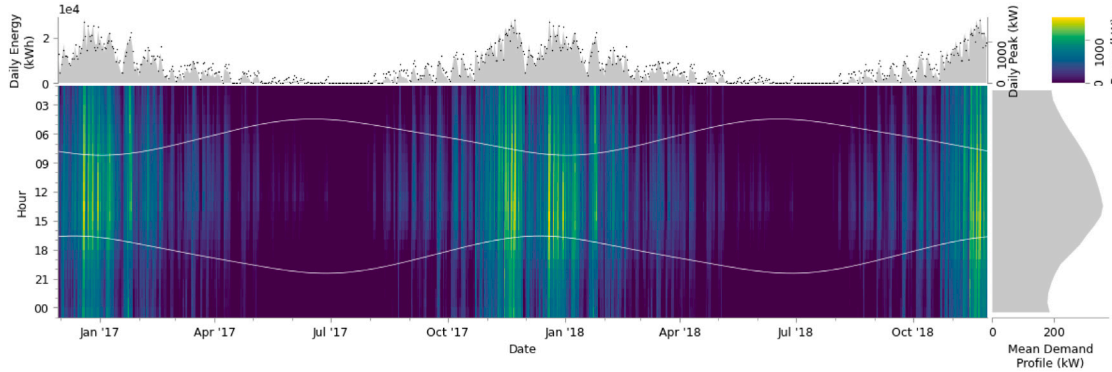
The primary concept of Graph Neural Networks (GNNs) revolves around integrating data associated with each node in a graph and its neighboring nodes, with the objective of utilizing the interdependencies and spatial connections between nodes. Instead of employing basic graph convolution techniques like [23], our approach implements a Gated Graph Neural Network (Gated GNN) [24]. Gated GNN utilizes a gated recurrent unit (GRU) as a recurrent function, simplifying the recurrence to a fixed number of steps.

The biggest modification from typical GNNs to Gated GNNs is the use of Gated Recurrent Units (GRU). The Gated GNN-filter also takes the edge type and edge direction into consideration. To this end, $e_{i,j}$ denotes the directed edge from node v_i to node v_j. The propagation process of recurrent-based Gated GNN can be summarized as follows [24]:

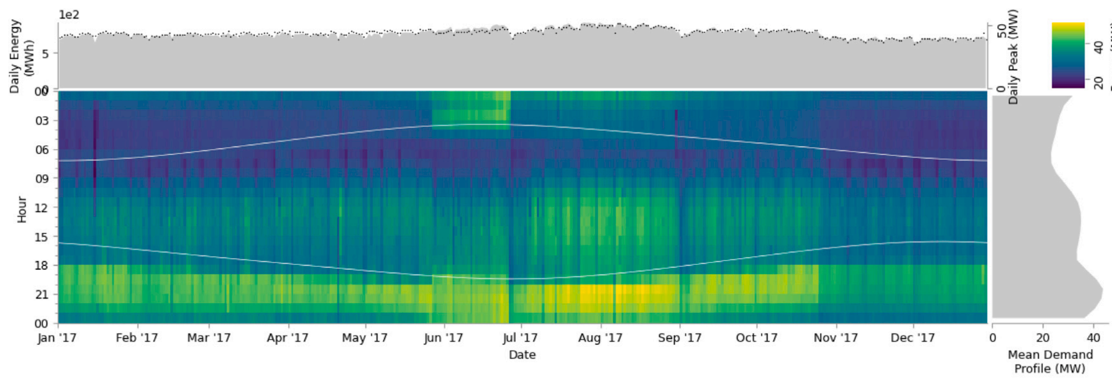
$$h_i^{(0)} = [x_i^T, 0]^T, \tag{4}$$

$$a_i^{(l)} = A_{i:}^T [h_1^{(l-1)} \cdots h_n^{(l-1)}]^T, \tag{5}$$

$$h_i^{(l)} = GRU(a_i^{(l)}, h_i^{(l-1)}), \tag{6}$$



(a) Heating demand of residential customer for two years.



(b) Electricity demand of zone 1 in Tetouan city for one year.

Fig. 3. Visualization of electricity demand for both use cases: (a) household level, (b) distribution level. Daily demand and the average demand profile are showed along the x- and y-axes, respectively. Here we also illustrate the times of sunrise and sunset on each day (as white lines in the heatmap), as these are related to energy demand for lighting and heating.

where $A_{i:} \in \mathbb{R}^{N\times 2N}$ are the two columns of blocks in A corresponding to node v_i. In Eq. (4), the initial node feature x_i are padded with extra zeros to make the input size equal to the hidden size. Eq. (5) computes $a_i^{(l)} \in \mathbb{R}^{2N}$ by aggregating information from different nodes via incoming and outgoing edges with parameters dependent on the edge type and direction. The following step uses a GRU unit to update the hidden state of node v by incorporating $a_i^{(l)}$ and the previous timestep hidden state $h_i^{(l-1)}$. This way, h_i corresponds to the output of the Gated GNN model. The gated GNN is effective at representing complex graph structures and is capable of enhancing GNN representational power.

3. Use cases

We evaluated the performance of our proposed X-CGNN approach in forecasting applications using two real-world power systems related datasets. To optimize the utility sector's utilization of our forecasting methods, it is essential to incorporate key utility data, particularly historical demand patterns. Furthermore, integrating external sources such as weather data improves the accuracy and reliability of our model. This comprehensive approach enables us to provide actionable insights for optimizing load forecasting, resource allocation, and grid management within the utility industry.

3.1. Use case 1: Household level electricity demand

The heating and electricity demand data are the results of an energy audit program aggregated for multiple load profiles of a residential customer. These profiles include HVAC systems loads, convenience power, elevator, etc. The datasets are gathered between December 2010 and November 2018 with a one-hour time step resolution. In addition to the historical energy demand values, data of weather variables are also available. The weather variables are air pressure, temperature, and humidity plus wind speed, and solar irradiation at the predetermined location. The data is publicly available [25]. Here we use the data for last two years, 2017 and 2018. Fig. 3 (a) shows the heating demand of this residential customer for these two years. In this figure daily demand and the average demand profile are showed along the x- and y-axes, respectively. Here we also illustrate the times of sunrise and sunset on each day (as white lines in the heatmap), as these are related to energy demand for lighting and heating. The upper insert shows the daily energy demand (gray area) and the daily peak demand (black dots). The right insert shows the average demand profile.

3.2. Use case 2: Distribution level electricity demand

Electric power demand data was collected from three different power distribution networks in Tetouan city, located in north Morocco. The data was obtained from the Supervisory Control and Data Acquisition System (SCADA) of Amendis, a public utility operator responsible for water and electricity distribution. The distribution network is powered by three source stations: Zone1: Quads, Zone2: Smir, and Zone3: Boussafou. The power demand data set covers the entire year of 2017 and was recorded every 10 min. Our visualization of Tetouan city's electricity demand within Zone 1 is shown in Fig. 3(b). The data is publicly available [26].

Various factors influence power demand, such as weather, income, population, and electricity prices. For this use case, we focused on utilizing weather features, including temperature, humidity, wind speed, diffuse flows, and general diffuse flows.

Table 1
Performance comparison of X-CGNN using different causal structure learning methods for distribution level electricity demand forecasting.

	30 min			60 min			90 min		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
RL [33]	650	940	3.98%	1045	1580	5.78%	1655	2167	9.80%
PNL [34]	605	898	3.53%	1056	1514	6.21%	1717	2367	9.61%
CORL [35]	584	883	3.39%	1037	1557	6.62%	1430	1980	**8.07**%
MCSL [36]	651	938	3.90%	1277	1783	7.38%	1569	2090	9.08%
DAG-GNN [37]	611	902	3.69%	1299	1768	7.47%	**1426**	**1952**	8.31%
NOTEARS [20]	**556**	**859**	**3.02%**	**1009**	**1535**	**5.25%**	1478	1997	8.49%

4. Results & discussions

4.1. Baseline methods

In this study, we compare the proposed X-CGNN model with the following existing models:

- RF: Random Forest model is an ensemble machine learning technique that combines multiple decision trees to make accurate predictions based on historical time-ordered data. It involves transforming time series datasets into a supervised learning problem [13].
- XGBoost: eXtreme gradient boosting (XGBoost), a scalable tree boosting model introduced by Chen and Guestrin [27], is a powerful machine learning algorithm renowned for its efficiency. It employs a gradient boosting framework that sequentially adds decision trees to iteratively refine predictions, offering robust performance.
- AR: An autoregressive model is when a value from a time series is regressed on previous values from that same time series [5].
- LSTM: Long Short-Term Memory, a recurrent neural network architecture capable of capturing long-term dependencies in sequential data, making it suitable for time series prediction [28]. LSTM effectively uses specialized memory cells, input gates, output gates, and forget gates to retain and utilize information from previous time steps for accurate predictions.
- STGCN: Spatiotemporal Graph Convolutional Networks, which integrate graph convolution into a 1D convolutional layer [29]. It constructs a graph using the physical distance between sensors, and the input to the GNN is a predefined graph.
- STEM: Spectral Temporal Graph Neural Networks, which leverage both inter-series correlation and temporal dependence in the spectral domain [30]. This model employs spectral analysis techniques to extract features from time-series data, and then graph neural networks are applied to capture relationships between data points in the latent correlation graph.
- MTGNN: Multivariate Time Series Graph Neural Network, consisting of three primary components: graph learning, graph convolution, and temporal convolution [31].
- Autoformer: This method utilizes a Transformer-based model with a decomposition architecture, incorporating an Auto-Correlation mechanism to effectively capture cross-temporal dependencies for forecasting [32].

4.2. Evaluation metrics

In order to assess the effectiveness of the proposed method, we utilize the following standard metrics to quantify the dissimilarity between the real value Y_t and the predicted value $\hat{Y}_t$.

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_t - \hat{Y}_t\right| \tag{7}$$

- Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Y_t - \hat{Y}_t\right)^2} \tag{8}$$

- Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_t - \hat{Y}_t}{Y_t}\right| \tag{9}$$

The units of MAE and RMSE are the same as those that were used for measuring. In case 1, where household level electricity demand data is being measured, the unit is KW while in case 2, distribution level electricity demand data is measured in MW. MAPE does not have a unit making it more useful when comparing datasets from different sources.

4.3. Experimental settings

The datasets were divided into a training set (60%), a validation set (20%), and a test set (20%). Independent models were trained to predict future time steps (horizons) of 1, 2, 3, 6 and 9. The training process involved 30 epochs, using the Adam optimizer with a learning rate of 0.001, an L2 regularization penalty of 0.0001, and a gradient clip of 5. For full reproducibility, implementation of X-CGNN on these two datasets can be found on GitHub.[1]

4.4. Causal methods and predictive performance analysis

To analyse the validity of our proposed model, we examine which causal model, which is derived from time series variables, is most appropriate for the application. In module 1 (causal graph discovery), we utilize various causal inference methods and calculate prediction results. In this study, we examine diverse methods for causal graph discovery, such as causal reinforcement learning-based methods such as RL [33], CORL [35], as well as other methods such as Post-NonLinear model (PNL) [34], the Masked Gradient-Based Method (MCSL) [36], DAG structure learning with graph neural networks (DAG-GNN) [37], and Continuous optimization for structure learning (NOTEARS) [20]. Table 1 illustrates the prediction results for distribution level electricity demand using different causal inference methods in X-CGNN framework. This analysis indicates that the DAG-GNN and NOTEARS models yield better results, making them more suitable for our application. In this paper, we use the causal model based on NOTEARS for the remaining analyses.

NOTEARS algorithm begins by initializing with a random matrix W representing the adjacency matrix of the graph. It then computes the gradient of the loss function with respect to W, followed by updating W using a specific update rule ensuring acyclicity, such as the soft-thresholding operator. Convergence is checked by monitoring changes in the loss function or the matrix W, and the steps of computing the gradient, updating W, and checking convergence are repeated iteratively until convergence criteria are met. Finally, post-processing

[1] Made available at publication time.

Table 2
Forecasting performance comparison of X-CGNN and other baseline models for use case 1, household power demand.

	1 h			2 h			3 h		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
RF [38]	99.89	100.89	44.01%	100.20	102.39	43.80%	101.02	104.65	43.60%
XGBoost [27]	88.17	103.12	27.08%	88.80	103.39	27.20%	89.99	113.51	26.09%
AR [5]	28.95	43.59	32.19%	64.09	93.59	43.17%	106.83	152.67	53.04%
STGCN [29]	34.19	57.67	17.04%	41.49	72.87	19.24%	46.03	83.2	20.62%
STEM [30]	50.99	87.8	22.71%	43.24	74.25	21.29%	51.8	93.85	22.71%
MTGNN [31]	22.75	31.23	10.23%	35.67	49.05	18.11%	44.52	62.22	24.02%
Autoformer [32]	45.31	75.04	16.82%	49.64	81.23	17.84%	53.1	87.24	**18.62%**
X-CGNN	**21.86**	**30.21**	**10.12%**	**32.41**	**45.88**	**15.65%**	**44.32**	**61.98**	20.3%

may be applied to refine the final structure, such as thresholding small edge weights to zero. This algorithm effectively utilizes continuous optimization techniques to learn the structure of directed acyclic graphs efficiently, without relying on discrete search procedures. By iteratively updating a matrix representation of the graph while enforcing acyclicity through a smooth penalty term, NOTEARS achieves accurate and scalable structure learning [20].

4.5. Forecasting results

We conducted forecasting experiments using data from the two aforementioned use cases in order to evaluate the performance of the proposed X-CGNN method. The results are compared with state-of-the-art methods, including RF [13], XGBoost [27], AR [5], LSTM [28], STGCN [29], STEM [30], MTGNN [31] and Autoformer [32]. We present the results of the prediction for household level power demand over three different time horizons in Table 2 and for distribution level in Table 3.

As demonstrated in Tables 2 and 3, RF, XGBoost, AR and LSTM models, which rely solely on historical time-series patterns, exhibit the lowest performance especially for long-term predictions.. While AR and LSTM do account for variables dependencies, these dependencies are not explicitly modeled. Based on the results, it is emphasized that it is important to consider these interdependencies among variables, which can be achieved using GNN-based methods such as STGCN, STEM, and MTGNN, as well as transformer-based methods such as Autoformer. According to Table 2 for household level prediction, proposed X-CGNN method has almost 53.6% improvement in RMSE and 55.9% improvement in MAE in comparison with LSTM for 2 h ahead prediction. Also X-CGNN compared to MTGNN method has almost 6.4% improvement in RMSE and 9.1% improvement in MAE for 2 h ahead prediction. And based on Table 3 for distribution level prediction, the proposed X-CGNN has almost 40.8% improvement in RMSE and 51.6% improvement in MAE compared with LSTM for 1 h ahead prediction. Also X-CGNN compared to MTGNN method has almost 7.8% improvement in RMSE and 16.5% improvement in MAE for 1 h ahead prediction. Furthermore, our model provides better interpretability, which we will examine and compare in detail in the next sections. It is worth mentioning that for use case 1, the MAPE values are relatively high. The reason for this is that during the summer season, the heat demand values are close to zero. Due to Eq. (9) of MAPE, the presence of small real values leads to higher error rates.

Figs. 4 and 5 provide a visual representation of the superior performance of X-CGNN in comparison with other methods for forecasting household and distribution power demand. Within Fig. 4, we exhibit the forecasting results for electricity and heat demand in usecase 1 over three consecutive days, employing the LSTM and X-CGNN methods. Also shown in Fig. 5 is the prediction of power demand for three different zones of Tetouan city (usecase 2) over three days using these two methods. Through these visual comparisons, we can discern the enhanced accuracy and predictive capabilities of the X-CGNN approach for electricity demand applications.

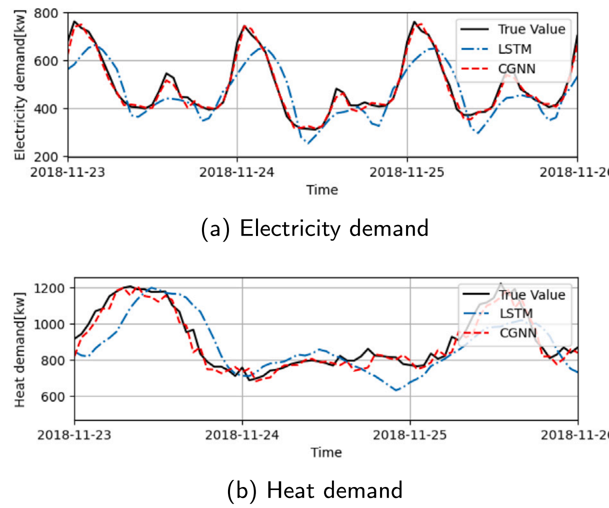


Fig. 4. Household level power demand forecasting comparison results for three days in November 2018: (a) Electricity demand; (b) Heat demand.

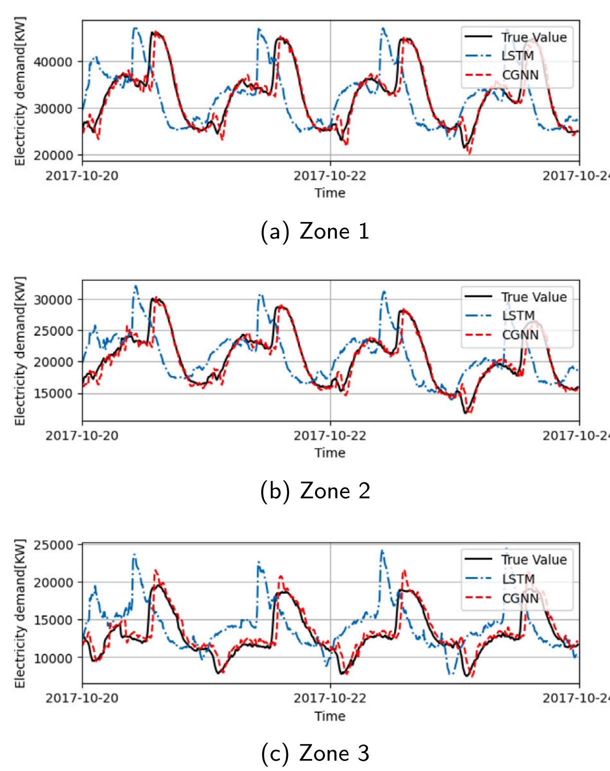


Fig. 5. Distribution level power demand forecasting comparison results for four days in October 2017: (a) zone 1; (b) Zone 2 ; (c) Zone 3.

Table 3
Forecasting performance comparison of X-CGNN and other baseline models for use case 2, distribution power demand.

	30 min			60 min			90 min		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
RF [38]	943	1371	5.60%	1624	2342	8.90%	2350	3315	13.01%
XGBoost [27]	1300	1688	7.70%	1797	2377	10.40%	2371	3091	13.60%
AR [5]	805	1118	3.76%	1528	2105	11.3%	2491	3335	11.13%
LSTM [28]	751	1008	4.23%	2085	2594	12.44%	3268	4260	20.48%
STGCN [29]	768	1121	4.1%	1347	1836	7.33%	1912	2470	10.59%
STEM [30]	732	1047	3.98%	1174	1733	6.2%	1791	2455	9.92%
MTGNN [31]	571	**857**	3.21%	1209	1666	7.35%	1548	2112	9.08%
Autoformer [32]	795	1359	10.1%	1035	1789	10.8%	**1298**	2202	11.52%
X-CGNN	**556**	859	**3.02%**	**1009**	**1535**	**5.25%**	1478	**1997**	**8.49%**

Table 4
Ablation study.

Method	X-CGNN	w/o CGD	w/o TC	w/o GNN
MAE	**1478**	1641	1878	1607
RMSE	**1997**	2287	2323	2222
MAPE	**8.49**	9.74	9.79	9.92

4.6. Ablation study

To further validate the effectiveness of our proposed X-CGNN, we conducted an ablation study to investigate the impact of each building block within the X-CGNN framework. We created variations of X-CGNN by removing individual modules, resulting in the following configurations:

- **w/o CGD**: X-CGNN without the causal graph discovery module. In this configuration, the causality graph is not used as an input to the GNN module; instead, the graph is learned.
- **w/o TC**: X-CGNN without the temporal convolution module. In this configuration, dilated convolution is not utilized, and temporal features are not extracted separately.
- **w/o GNN**: X-CGNN without the GNN module. In this configuration, GNN module is not taken into account, resulting in a model solely reliant on temporal convolution without modeling and considering the connections between variables.

Table 4 presents the forecasting errors (with three error metrics) with time horizon 9 (90 min) for Use Case 2.

The ablation analysis allows us to understand the significance of each module in our model. The results in Table 4 indicate that all three modules within the X-CGNN architecture are crucial for achieving high predictive performance. For instance, the GNN module enhances the model by approximately 8%, the TC module exhibits a substantial 21% improvement, and the CGD module contributes to a notable 9.9% enhancement, as assessed by the MAE metric. Removing any individual module significantly impairs the accuracy and effectiveness of the model. This highlights the importance of each component in the X-CGNN framework for successful forecasting.

4.7. Explainability results

Embedding causal relationships within Graph Neural Networks (GNNs) notably enhances their explainability. Here we evaluate both the global and local explanations of the proposed model.

4.7.1. Global explanations

We provide a global explanation of the method by examining causal graphs and explaining their interdependencies, as well as comparing it to the methods by which they learn the graph during training.

In regard to use case 1, household level electricity demand, Fig. 6 illustrates the extracted graphs using X-CGNN and MTGNN. Fig. 6(a) shows the graph extracted from MTGNN method, while Fig. 6(b) indicates the causality graph derived from the proposed X-CGNN. In addition to its denseness, complexity, and computational complexity, the MTGNN graph is difficult to interpret. By contrast, the causality graph derived from X-CGNN is more interpretable and computation-efficient.

There are some directed edges shown in Fig. 6(a) that are marked with red color. These are incorrect connections between variables, such as electricity demand having direct influence on wind speed. Consequently, MTGNN's performance has been affected by these incorrect connections. Nonetheless, in causal graph 6(b), we have marked these connections with green color to denote the correct direction in our proposed method. Experts hold the view that these connections are precise, and this causal graph offers a comprehensive comprehension of the model's functioning, contributing to its superior performance.

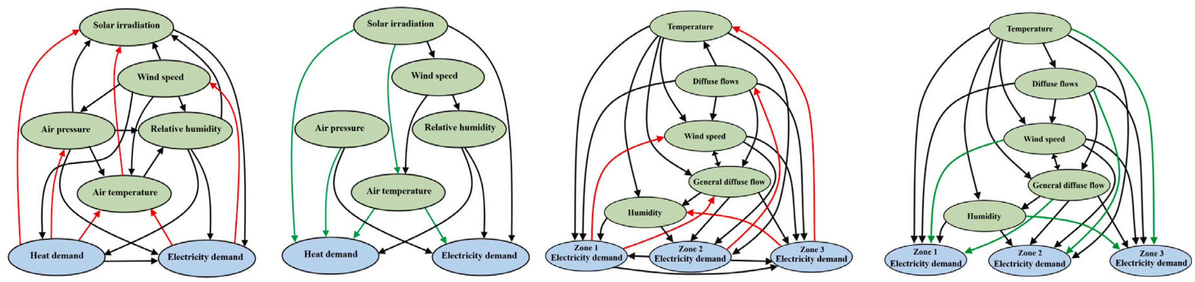
Similarly, in the context of use case 2, electricity demand in Tetouan city, the clear advantage of X-CGNN becomes evident. For the MTGNN graph, incorrectly connected edges are highlighted in red on Fig. 6(c). A noteworthy example is the incorrect assertion that zone 3's energy consumption affects temperature, which has been debunked. By contrast, Fig. 6(d) illustrates the causal graph form X-CGNN model, which captures the causal interdependent nature of variables. It should be noted that the causal graph is not ideal and may have connections that are difficult to interpret. However, overall, it offers much greater interpretability and has contributed to better performance of the model. Nevertheless, it's crucial to emphasize that understanding the causal graph often requires expert knowledge, as it involves intricate relationships and dependencies within the data.

4.7.2. Local explanations

For the purpose of obtaining local explanations and specific insights into the critical time lags and significant features for electricity demand prediction, post-hoc interpretability methods have been employed. Both qualitative and quantitative analyses are performed.

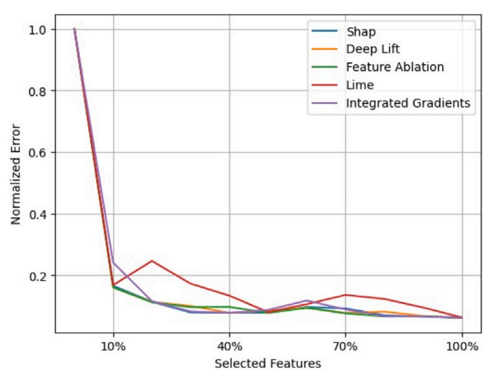
Fig. 7 presents a comparative assessment of various XAI techniques. Notably, by leveraging crucial features from each input sequence length, the top 'k' features are selected and tested. The decreasing error curve is obtained by varying k values from k=[10%, …, 100%] within a test set. This analysis effectively identifies which method identifies significant features. Fig. 7 depicts the performance of XAI methods. Comparisons with numerous baselines, including SHAP [39], DeepLift [40], Feature Ablation [41], LIME [42] and Integrated Gradients [43], demonstrate that the LIME method exhibits the poor performance, whereas the feature ablation method excels in discerning crucial features and time steps. Continuing, we proceed with a thorough examination of the outcomes from the superior model, specifically focusing on the Feature Ablation technique.

Figs. 8 and 9 present the results related to the essential features and time steps for prediction using the X-CGNN model for two use cases, employing the post-hoc Feature Ablation method. The depicted heatmaps effectively illustrate the significance of each feature and time step. Fig. 8(a) displays the output of the feature ablation for household electricity demand. For better visualization, we have performed an adjusted scale in Fig. 8(b). For this illustration, we have sequentially

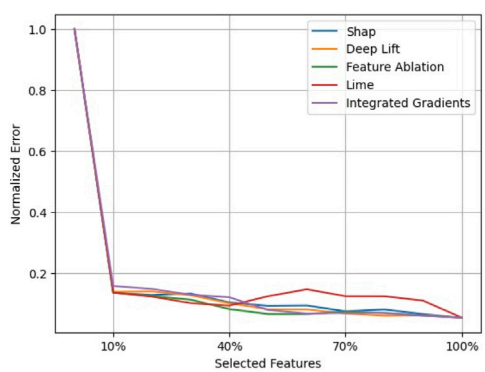


(a) Learned graph for household level use case. (b) Causal graph for household level use case. (c) Learned graph for regional level use case. (d) Causal graph for regional level use case.

Fig. 6. Visual representation of the learned graph in MTGNN and the inferred causal graph by the proposed X-CGNN method. It is indicated that red edges are incorrect connections while green edges are correct causal connections. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



(a) Household level electricity demand.



(b) Distribution level electricity demand.

Fig. 7. Quantitative analysis of XAI methods. This curves illustrates the normalized error magnitude of prediction, taking into account the percentage of important features identified through various XAI methods. For example, if we extract the most critical features using specified methods and make predictions using only 10% of the critical features, the error significantly decreases. This indicates the accurate identification of influential features in prediction. Therefore, the lower the curve, the higher the method's accuracy.

assigned numerical values in ascending order, starting from the least importance and progressing to the most important.

As depicted in Fig. 8, our prediction model initially looks at the previous hour, followed by the preceding hours of the previous day, around 24 h earlier, and to a certain extent, two days prior. This model behavior, emphasizing past data for future prediction, seems entirely logical, especially when considering household consumption. This alignment between interpretability insights and domain expertise significantly bolsters our confidence in the model.

Now, we aim to examine the impact of these results on selecting the appropriate sequence length for prediction. As evident in Fig. 8, our proposed model assigns greater importance to three specific time-frames: the previous hours, approximately similar hours of the previous day, and similarly, the day before that.

Table 5 displays the performance comparison of X-CGNN with various input sequence lengths for predicting household electricity demand. The results accurately recognize the importance of temporal intervals for prediction as indicated by explainability insights.

For distribution level electricity demand, Fig. 9 indicates that the proposed model assigns significant importance to data up to approximately 1.5 h prior. After this, the significance of time steps decreases. These findings were investigated considering various sequence lengths, as depicted in Table 6. The findings suggest that for short-term predictions (e.g., 30 min), one hour of preceding data is sufficient and provides better predictive performance. However, for longer-term forecasts such as 1 h and 1.5 h, a sequence length of 180 min proves to be more suitable. This analysis not only allows us to understand the model but also enables the design of the model using interpretable methods, resulting in reduced error even with less data. It provides us with a lightweight model suitable for real-time applications when implementing the model.

4.8. Computation time

We evaluate the computational cost of X-CGNN against two recent competitive methods, MTGNN [31] and Autoformer [32], as presented in Table 7. Computational running times are computed separately for training per epoch and inference for each forecasting point. The findings demonstrate that Autoformer exhibits the lowest training time for both use cases, followed by the proposed X-CGNN model, while MTGNN necessitates a significantly longer training duration. However, X-CGNN showcases the lowest inference time for both use cases. Taking into account the satisfactory performance of our proposed method in prediction, its computation cost is also deemed suitable compared to recent approaches.

4.9. Practical consideration and challenges

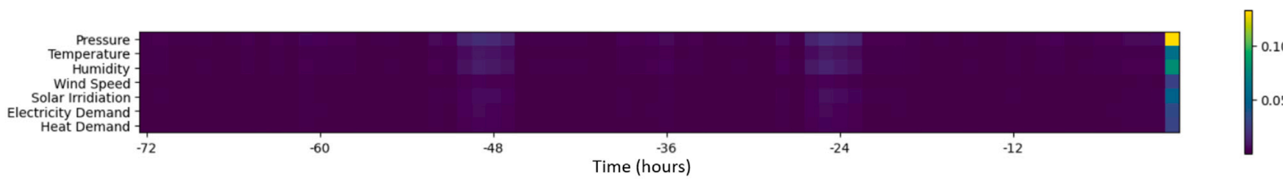
Electricity demand forecasting is complex, demanding a comprehensive approach to data collection and analysis. We encounter practical challenges, notably the need for both internal and external data sources. While historical demand records provide insights into past consumption patterns, external factors like weather and economic activity significantly influence electricity demand, enhancing forecasting accuracy when incorporated. However, accessing comprehensive datasets including both internal and external variables is often difficult. Additionally, understanding power grid topology is crucial but obtaining detailed and up-to-date data presents challenges due to sensitivity and privacy concerns. Balancing privacy with data access is crucial, requiring secure protocols or alternative sources. Also it's important to note that in the energy domain, uncertainty plays a vital role, and

Table 5
Performance comparison of X-CGNN with various input sequence lengths for the prediction of household electricity demand. According to the results obtained by feature ablation, sequence lengths are selected.

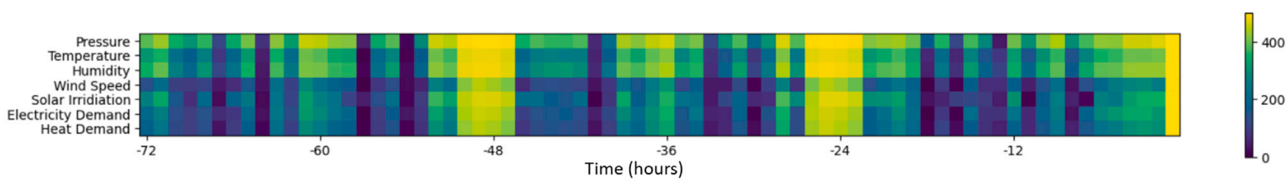
	1 h			2 h			3 h		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
X-CGNN-6 h	33.22	44.12	13.21%	62.33	83.32	25.11%	89.08	118.75	36.91%
X-CGNN-30 h	22.08	31.37	13.15%	35.94	51.38	23.47%	47.34	69.06	22.25%
X-CGNN-72 h	**21.86**	**30.21**	**10.12%**	**32.41**	**45.88**	**15.65%**	**44.32**	**61.98**	**20.3%**

Table 6
Performance comparison of X-CGNN with various input sequence lengths for the prediction of distribution level electricity demand. According to the results obtained by feature ablation, sequence lengths are selected.

	30 min			60 min			90 min		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
X-CGNN-30 min	2251	2926	10.34%	1772	2423	8.76%	1667	2400	**8.12%**
X-CGNN-60 min	**544**	**847**	**2.91%**	1110	1606	6.17%	1604	2170	9.47%
X-CGNN-180 min	556	859	3.02%	**1009**	**1535**	**5.25%**	**1478**	**1997**	8.49%

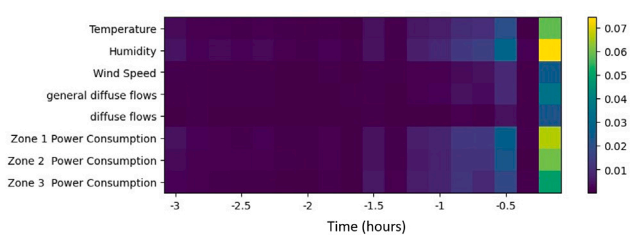


(a) Heatmap of the importance of variables and previous time steps.

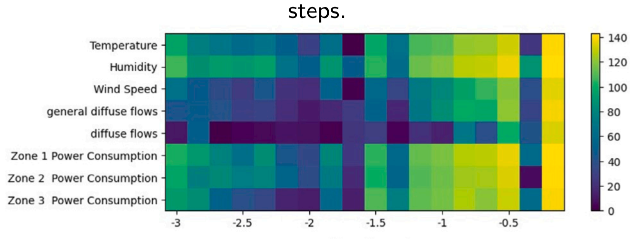


(b) Adjusted scale heatmap of the importance of variables and previous time steps.

Fig. 8. Visualization of the importance of variables and previous time steps for the household level electricity demand forecasting.



(a) Heatmap of the importance of variables and previous time steps.



(b) Adjusted scale heatmap of the importance of variables and previous time steps.

Fig. 9. Visualization of the importance of variables and previous time steps for the distribution level electricity demand forecasting.

Table 7
Computational running times.

	Training time (s)		Inference time (ms)	
	Use case 1	Use case 2	Use case 1	Use case 2
MTGNN [31]	33.25	153.42	1.332	0.289
Autoformer [32]	14.56	38.01	0.906	0.409
X-CGNN	22.49	85.78	0.109	0.167

our current approach has limitations in probabilistic forecasting. Therefore, for future endeavors, we aim to integrate uncertainty to achieve highly accurate probabilistic predictions, thus making the model more reliable.

5. Conclusion

In summary, the pressing need for reliable electricity demand forecasting in the evolving landscape of renewable energy integration demands models that are not only advanced but also explainable. Our study introduces the eXplainable Causal Graph Neural Network (X-CGNN) as a solution to address the limitations of existing black-box models in electricity demand forecasting. The X-CGNN method, employing intrinsic and global causal inferences alongside local post-hoc analyses, demonstrates superior interpretability and state-of-the-art performance. Through rigorous validation on real-world datasets spanning both household and distribution levels, our approach proves its efficacy, providing valuable insights into inter-variable dependencies. This advancement not only enriches forecasting accuracy but also

Table 8
Table of hyperparameters.

Model	Number of layers	Activation function	Loss function	Learning rate	Epoches	Dropout rate	Optimizer
LSTM [28]	4	Tanh	Huber	0.0001	30	–	Adam
STGCN [29]	2	ReLU	L2Loss	0.0001	30	–	RMSProp
STEM [30]	5	LeakyReLU	mse	0.0001	30	0.5	RMSProp
MTGNN [31]	5	ReLU	L1Loss	0.0001	30	0.3	Adam
Autoformer [32]	3	GELU	mse	0.0001	30	0.05	Adam

fosters a deeper understanding of the complexities within multivariate time series forecasting for electricity demand, thereby contributing significantly to the advancement of the power system technologies.

CRediT authorship contribution statement

Amir Miraki: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology. **Pekka Parviainen:** Writing – review & editing, Supervision, Methodology. **Reza Arghandeh:** Writing – review & editing, Writing – original draft, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The two datasets used in this study are publically available.

Appendix

In this section, we provide the details of our implementation for reproducibility. In our experiment, for the X-CGNN and MTGNN methods, the "weight decay rate" is set to 0.00001, the "graph convolution depth" is set to 2, "k" is set to 8, convolution and residual channels are set to 16, "skip channels" are set to 32, and "end channels" are set to 64. For the STEM method, the "decay rate" is set to 0.5, the "leaky rate" is set to 0.5, and the "exponential decay step" is set to 1. In the STGCN method, the "order of Chebyshev polynomial" is set to 3, and the "decay rate" is set to 0.7. In the Autoformer method, the "decoder input size" is set to 9, the "dimension of the model" is set to 128, the "window size of moving average" is set to 25, and the "number of heads" is set to 8. It consists of 2 layers as encoders and 2 layers as decoders in the LSTM method. Each LSTM layer has 100 neurons, and "return sequence" is set to True. In the first encoder, the "layer return" state is set to True. Additional hyperparameters are presented in Table 8.

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